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FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

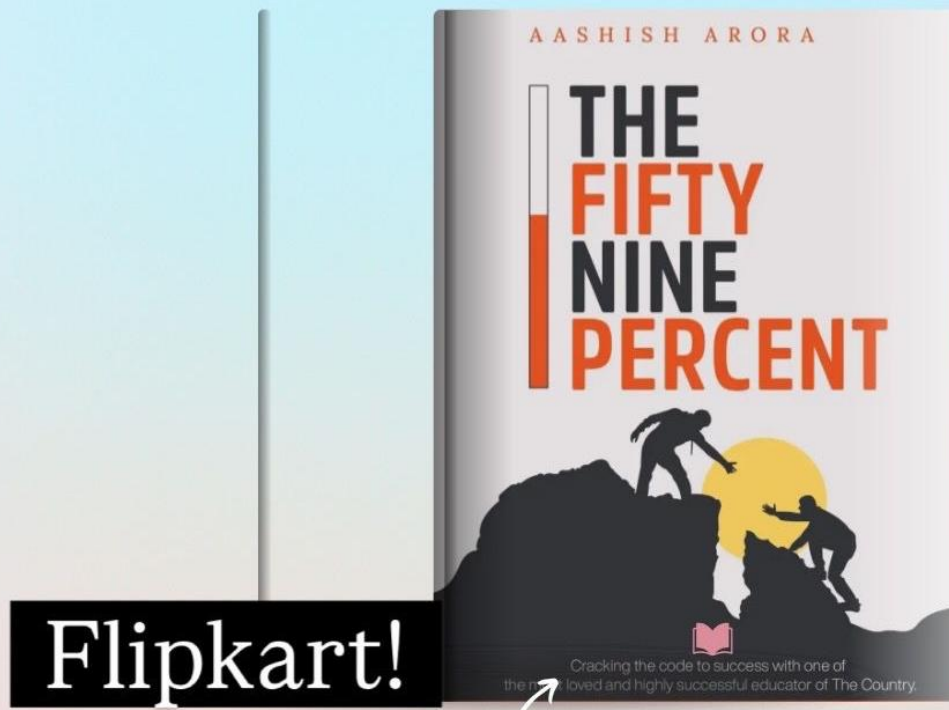


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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $48\% \text{ of } 1375 + \frac{7}{11} \text{ of } 792 - \sqrt{5776} = ? \times 8$

- A. 128
- B. 132
- C. 136
- D. 140
- E. None of these

2. $? \% \text{ of } 960 + \frac{13}{16} \text{ of } 320 = \frac{5}{6} \text{ of } 1080 + 2^7$

- A. 80
- B. 70
- C. 75
- D. 85
- E. None of these

3. $(\sqrt{6084} \times 14) - (15 \times \sqrt[3]{2197}) + 31.25\% \text{ of } 640 = ? + \frac{1}{8} \text{ of } 960$

- A. 957
- B. 967

C. 987

D. 977

E. None of these

4. $2\frac{1}{5} \text{ of } 250 + 75\% \text{ of } 384 - 16^2 = ? \times 6$

- A. 93
- B. 97
- C. 103
- D. 107
- E. None of these

5. $\frac{11}{12} \text{ of } 720 - 44.44\% \text{ of } 810 + 3^5 = ? + \sqrt{3136}$

- A. 487
- B. 477
- C. 497
- D. 507
- E. None of these

6. $[\frac{9}{16} \text{ of } 1088 + 18.75\% \text{ of } 1536] \div 8 + 7^2 = ?$

- A. 157.5
- B. 159.5

C. 163.5

D. 161.5

E. None of these

$$7. ? \times 18 - 68.75\% \text{ of } 512 = \frac{5}{7} \text{ of } 980 + 4^3$$

A. 58

B. 62

C. 66

D. 70

E. None of these

$$8. (23^2 - 17^2) \div 5 + 32.5\% \text{ of } 800 + \frac{1}{15} \text{ of } 450 = ?$$

A. 326

B. 332

C. 338

D. 344

E. None of these

$$9. 7.69\% \text{ of } 2730 + 86.66\% \text{ of } 750 - 12.5\% \text{ of } 1376 = ? \times 8$$

A. 78

B. 82

C. 90

D. 86

E. None of these

$$10. \sqrt[3]{175616} + \sqrt{5184} + 41.66\% \text{ of } 1008 = ? + 5^3$$

A. 413

B. 423

C. 433

D. 443

E. None of these

$$11. \frac{3}{8} \text{ of } 640 + \frac{5}{9} \text{ of } 972 - 14.28\% \text{ of } 700 = ? + 2^5$$

A. 648

B. 638

C. 658

D. 668

E. None of these

$$12. [(24 \times 36) - (18 \times 28)] \div 12 + 83.33\% \text{ of } 432 = ? \div 3$$

A. 1110

B. 1140

C. 1200

D. 1170

E. None of these

$$13. ? - 56.25\% \text{ of } 896 = 22^2 + 31.25\% \text{ of } 768$$

A. 1208

B. 1218

C. 1228

D. 1238

E. None of these

$$14. \frac{14}{15} \text{ of } 600 + \frac{2}{11} \text{ of } 968 - \sqrt{1296} = ? \times 7$$

A. 100

B. 96

C. 104

D. 108

E. None of these

$$15. (15^2 + 21^2 - 18^2) + 6.25\% \text{ of } 2048 = ? + 37.5\% \text{ of } 384$$

A. 316

B. 326

C. 336

D. 346

E. None of these

$$16. 5\frac{2}{7} \text{ of } 133 + 27.27\% \text{ of } 693 - 4^4 = ? \times 6$$

A. 98

B. 102

C. 106

D. 110

E. None of these

$$17. \sqrt{?} + 37.5\% \text{ of } 768 = \frac{5}{6} \text{ of } 720 + 12 \times 22$$

A. 321489

B. 331776

C. 341056

D. 351649

E. None of these

$$18. 25\% \text{ of } \{18 \times 64 - 12 \times 36\} + 45.45\% \text{ of } 726 = ?$$

A. 490

B. 500

C. 520

D. 510

E. None of these

$$19. 72.72\% \text{ of } 1210 - 58.33\% \text{ of } 1056 + ? = 17^2 + 19^2$$

A. 376

B. 386

C. 396

D. 406

E. None of these

$$20. \left[\frac{11}{16} \text{ of } 640 + 93.33\% \text{ of } 450 \right] \div ? = \sqrt[3]{27000} + 13$$

- A. 16
B. 18
C. 20
D. 22
E. None of these

Answers:

1. C
2. A
3. D
4. B
5. A
6. D
7. B
8. C
9. D
10. B
11. A
12. D
13. C
14. A
15. B
16. C
17. B
18. D
19. B

20. C

Solutions:

$$1. 48\% \text{ of } 1375 + \frac{7}{11} \text{ of } 792 - \sqrt{5776} = ? \times 8$$

$$48\% \text{ of } 1375 = 660, \frac{7}{11} \text{ of } 792 = 504 \text{ and } \sqrt{5776} = 76$$

$$660 + 504 - 76 = ? \times 8$$

$$1088 = ? \times 8$$

$$? = 136$$

$$2. ?\% \text{ of } 960 + \frac{13}{16} \text{ of } 320 = \frac{5}{6} \text{ of } 1080 + 2^7$$

$$\frac{13}{16} \text{ of } 320 = 260, \frac{5}{6} \text{ of } 1080 = 900 \text{ and } 2^7 = 128$$

$$?\% \text{ of } 960 + 260 = 900 + 128$$

$$?\% \text{ of } 960 = 1028 - 260 = 768$$

$$? = 768 \times 100 \div 960 = 80$$

$$3. (\sqrt{6084} \times 14) - (15 \times \sqrt[3]{2197}) + 31.25\% \text{ of } 640 = ? + \frac{1}{8} \text{ of } 960$$

$$\sqrt{6084} = 78, \sqrt[3]{2197} = 13, 31.25\% = \frac{5}{16} \text{ and } \frac{1}{8} \text{ of } 960 = 120$$

$$78 \times 14 - 15 \times 13 + \frac{5}{16} \times 640 = ? + 120$$

$$1092 - 195 + 200 = ? + 120$$

$$1097 - 120 = ?$$

$$? = 977$$

$$4. 2\frac{1}{5} \text{ of } 250 + 75\% \text{ of } 384 - 16^2 = ? \times 6$$

$$2\frac{1}{5} = \frac{11}{5}, 75\% = \frac{3}{4} \text{ and } 16^2 = 256$$

$$\frac{11}{5} \times 250 + \frac{3}{4} \times 384 - 256 = ? \times 6$$

$$550 + 288 - 256 = ? \times 6$$

$$582 = ? \times 6$$

$$? = 97$$

$$5. \frac{11}{12} \text{ of } 720 - 44.44\% \text{ of } 810 + 3^5 = ? + \sqrt{3136}$$

$$\frac{11}{12} \text{ of } 720 = 660, 44.44\% = \frac{4}{9}, 3^5 = 243 \text{ and } \sqrt{3136} = 56$$

$$660 - \frac{4}{9} \times 810 + 243 = ? + 56$$

$$660 - 360 + 243 = ? + 56$$

$$543 = ? + 56$$

$$? = 487$$

$$6. \left[\frac{9}{16} \text{ of } 1088 + 18.75\% \text{ of } 1536 \right] \div 8 + 7^2 = ?$$

$$\frac{9}{16} \text{ of } 1088 = 612, 18.75\% = \frac{3}{16} \text{ and } 7^2 = 49$$

$$\left[612 + \frac{3}{16} \times 1536 \right] \div 8 + 49 = ?$$

$$[612 + 288] \div 8 + 49 = ?$$

$$900 \div 8 + 49 = ?$$

$$112.5 + 49 = 161.5$$

$$7. ? \times 18 - 68.75\% \text{ of } 512 = \frac{5}{7} \text{ of } 980 + 4^3$$

$$68.75\% = \frac{11}{16}, \frac{5}{7} \text{ of } 980 = 700 \text{ and } 4^3 = 64$$

$$? \times 18 - \frac{11}{16} \times 512 = 700 + 64$$

$$? \times 18 - 352 = 764$$

$$? \times 18 = 1116$$

$$? = 62$$

$$8. (23^2 - 17^2) \div 5 + 32.5\% \text{ of } 800 + \frac{1}{15} \text{ of } 450 = ?$$

$$23^2 = 529, 17^2 = 289, 32.5\% = \frac{13}{40} \text{ and } \frac{1}{15} \text{ of } 450 = 30$$

$$(529 - 289) \div 5 + \frac{13}{40} \times 800 + 30 = ?$$

$$240 \div 5 + 260 + 30 = ?$$

$$48 + 260 + 30 = 338$$

$$9. 7.69\% \text{ of } 2730 + 86.66\% \text{ of } 750 - 12.5\% \text{ of } 1376 = ? \times 8$$

$$7.69\% = \frac{1}{13}, 86.66\% = \frac{13}{15} \text{ and } 12.5\% = \frac{1}{8}$$

$$\frac{1}{13} \times 2730 + \frac{13}{15} \times 750 - \frac{1}{8} \times 1376 = ? \times 8$$

$$210 + 650 - 172 = ? \times 8$$

$$688 = ? \times 8$$

$$? = 86$$

$$10. \sqrt[3]{175616} + \sqrt{5184} + 41.66\% \text{ of } 1008 = ? + 5^3$$

$$\sqrt[3]{175616} = 56, \sqrt{5184} = 72, 41.66\% = \frac{5}{12} \text{ and } 5^3 = 125$$

$$56 + 72 + \frac{5}{12} \times 1008 = ? + 125$$

$$56 + 72 + 420 = ? + 125$$

$$548 = ? + 125$$

$$? = 423$$

$$11. \frac{3}{8} \text{ of } 640 + \frac{5}{9} \text{ of } 972 - 14.28\% \text{ of } 700 = ? + 2^5$$

$$\frac{3}{8} \text{ of } 640 = 240, \frac{5}{9} \text{ of } 972 = 540, 14.28\% = \frac{1}{7} \text{ and } 2^5 = 32$$

$$240 + 540 - \frac{1}{7} \times 700 = ? + 32$$

$$240 + 540 - 100 = ? + 32$$

$$680 = ? + 32$$

$$? = 648$$

$$12. [(24 \times 36) - (18 \times 28)] \div 12 + 83.33\% \text{ of } 432 = ? \div 3$$

$$24 \times 36 = 864, 18 \times 28 = 504 \text{ and } 83.33\% = \frac{5}{6}$$

$$[864 - 504] \div 12 + \frac{5}{6} \times 432 = ? \div 3$$

$$360 \div 12 + 360 = ? \div 3$$

$$30 + 360 = ? \div 3$$

$$390 = ? \div 3$$

$$? = 1170$$

$$13. ? - 56.25\% \text{ of } 896 = 22^2 + 31.25\% \text{ of } 768$$

$$56.25\% = \frac{9}{16}, 22^2 = 484 \text{ and } 31.25\% = \frac{5}{16}$$

$$? - \frac{9}{16} \times 896 = 484 + \frac{5}{16} \times 768$$

$$? - 504 = 484 + 240$$

$$? - 504 = 724$$

$$? = 1228$$

$$14. \frac{14}{15} \text{ of } 600 + \frac{2}{11} \text{ of } 968 - \sqrt{1296} = ? \times 7$$

$$\frac{14}{15} \text{ of } 600 = 560, \frac{2}{11} \text{ of } 968 = 176 \text{ and } \sqrt{1296} = 36$$

$$560 + 176 - 36 = ? \times 7$$

$$700 = ? \times 7$$

$$? = 100$$

$$15. (15^2 + 21^2 - 18^2) + 6.25\% \text{ of } 2048 = ? + 37.5\% \text{ of } 384$$

$$15^2 = 225, 21^2 = 441, 18^2 = 324, 6.25\% = \frac{1}{16} \text{ and } 37.5\% = \frac{3}{8}$$

$$(225 + 441 - 324) + \frac{1}{16} \times 2048 = ? + \frac{3}{8} \times 384$$

$$342 + 128 = ? + 144$$

$$470 = ? + 144$$

$$? = 326$$

$$16. 5\frac{2}{7} \text{ of } 133 + 27.27\% \text{ of } 693 - 4^4 = ? \times 6$$

$$5\frac{2}{7} = \frac{37}{7}, 27.27\% = \frac{3}{11} \text{ and } 4^4 = 256$$

$$\frac{37}{7} \times 133 + \frac{3}{11} \times 693 - 256 = ? \times 6$$

$$703 + 189 - 256 = ? \times 6$$

$$636 = ? \times 6$$

$$? = 106$$

$$17. \sqrt{?} + 37.5\% \text{ of } 768 = \frac{5}{6} \text{ of } 720 + 12 \times 22$$

$$37.5\% = \frac{3}{8}, \frac{5}{6} \text{ of } 720 = 600 \text{ and } 12 \times 22 = 264$$

$$\sqrt{?} + \frac{3}{8} \times 768 = 600 + 264$$

$$\sqrt{?} + 288 = 864$$

$$\sqrt{?} = 576$$

$$? = 576^2 = 331776$$

$$18. 25\% \text{ of } \{18 \times 64 - 12 \times 36\} + 45.45\% \text{ of } 726 = ?$$

$$25\% = \frac{1}{4}, 18 \times 64 = 1152, 12 \times 36 = 432 \text{ and } 45.45\% = \frac{5}{11}$$

$$\frac{1}{4} \times \{1152 - 432\} + \frac{5}{11} \times 726 = ?$$

$$\frac{1}{4} \times 720 + 330 = ?$$

$$180 + 330 = 510$$

$$19. 72.72\% \text{ of } 1210 - 58.33\% \text{ of } 1056 + ? = 17^2 + 19^2$$

$$72.72\% = \frac{8}{11}, 58.33\% = \frac{7}{12}, 17^2 = 289 \text{ and } 19^2 = 361$$

$$\frac{8}{11} \times 1210 - \frac{7}{12} \times 1056 + ? = 289 + 361$$

$$880 - 616 + ? = 650$$

$$264 + ? = 650$$

$$? = 386$$

$$20. [\frac{11}{16} \text{ of } 640 + 93.33\% \text{ of } 450] \div ? = \sqrt[3]{27000} + 13$$

$$\frac{11}{16} \text{ of } 640 = 440, 93.33\% = \frac{14}{15} \text{ and } \sqrt[3]{27000} = 30$$

$$[440 + \frac{14}{15} \times 450] \div ? = 30 + 13$$

$$[440 + 420] \div ? = 43$$

$$860 \div ? = 43$$

$$? = 20$$

2. ARITHMETIC QUESTIONS

Q1: There are two companies P and Q. The number of male employees in both companies is the same. The number of female employees in company Q is 40% less than that in company P. If the number of female employees in company P increases by 20% and the number of male employees in company P decreases by 40%, then the total number of employees in both companies becomes equal. Find the initial ratio of total employees in company P to company Q.

दो कंपनियाँ P और Q हैं। दोनों कंपनियों में पुरुष कर्मचारियों की संख्या समान है। कंपनी Q में महिला कर्मचारियों की संख्या कंपनी P की तुलना में 40% कम है। यदि कंपनी P में महिला कर्मचारियों की संख्या 20% बढ़ जाती है और पुरुष कर्मचारियों की संख्या 40% घट जाती है, तो दोनों कंपनियों में कुल कर्मचारियों की संख्या समान हो जाती है। कंपनी P और कंपनी Q के कुल कर्मचारियों का प्रारंभिक अनुपात ज्ञात कीजिए।

- (A) 5:3
- (B) 21:25
- (C) 6:5
- (D) 25:21
- (E) 7:6

Q2: The average consumption of tea in an office is 210 cups per working day, while the average monthly consumption of tea per employee is 35 cups. If there are 264 working days in a year, find the number of employees in the office.

एक कार्यालय में चाय की औसत खपत 210 कप प्रति कार्य दिवस है, जबकि प्रति कर्मचारी चाय की औसत मासिक खपत 35 कप है। यदि एक वर्ष में 264 कार्य दिवस हैं, तो कार्यालय में कर्मचारियों की संख्या ज्ञात कीजिए।

- (A) 120
- (B) 132
- (C) 144
- (D) 126
- (E) 156

Q3: A van started from city A and reached city B at a speed of X km/hr. A car started from city B and reached city A at a speed of $(X + 30)$ km/hr. If the car takes 3 hours less than the van to complete the journey, find the value of $X + 8$. The

distance between city A and city B is 540 km.

एक वैन शहर A से चली और X km/hr की गति से शहर B पहुँची। एक कार शहर B से चली और $(X + 30)$ km/hr की गति से शहर A पहुँची। यदि कार यात्रा पूरी करने में वैन से 3 घंटे कम समय लेती है, तो $X + 8$ का मान ज्ञात कीजिए। शहर A और शहर B के बीच की दूरी 540 किमी है।

- (A) 56
- (B) 70
- (C) 65
- (D) 75
- (E) 68

Q4: The average of five consecutive numbers that are multiples of 10 is 70. The difference between the highest number and the second smallest number is equal to the largest even number in a set of four consecutive even numbers. Find the square of the smallest even number of that set.

10 के पाँच क्रमागत गुणजों का औसत 70 है। सबसे बड़ी संख्या और दूसरी सबसे छोटी संख्या का अंतर चार क्रमागत सम संख्याओं के एक समूह की सबसे बड़ी सम संख्या के बराबर है। उस समूह की सबसे छोटी सम संख्या का वर्ग ज्ञात कीजिए।

- (A) 484
- (B) 576
- (C) 400
- (D) 625
- (E) 676

Q5: The perimeter of a right-angled triangle is 84 meters and its hypotenuse is 35 meters. If the base is the smallest side, find the length of the base.

एक समकोण त्रिभुज का परिमाप 84 मीटर है और उसका कर्ण 35 मीटर है। यदि आधार सबसे छोटी भुजा है, तो आधार की लंबाई ज्ञात कीजिए।

- (A) 15 meters
- (B) 18 meters
- (C) 24 meters
- (D) 21 meters
- (E) 28 meters

Q6: The average salary of Kiran and Megha is Rs. 4500. If Kiran's salary is increased by 40% and Megha's salary is increased by 25%, then their total salary increases by 30%. Find how much percent more Megha's initial salary is than

Kiran's initial salary.

किरण और मेघा की औसत सैलरी Rs. 4500 है। यदि किरण की सैलरी में 40% और मेघा की सैलरी में 25% की वृद्धि होती है, तो उनकी कुल सैलरी में 30% की वृद्धि होती है। ज्ञात कीजिए कि मेघा की प्रारंभिक सैलरी, किरण की प्रारंभिक सैलरी से कितने प्रतिशत अधिक है?

- (A) 50%
- (B) 75%
- (C) 100%
- (D) 150%
- (E) 200%

Q7: The average weight of a group is 45 kg. If a person weighing 63 kg leaves the group, then the average weight of the group decreases by 2 kg. Find the number of people in the group initially.

एक समूह का औसत वजन 45 किग्रा है। यदि 63 किग्रा वजन वाला एक व्यक्ति समूह छोड़ देता है, तो समूह का औसत वजन 2 किग्रा कम हो जाता है। प्रारंभ में समूह में लोगों की संख्या ज्ञात कीजिए।

- (A) 10
- (B) 12
- (C) 14
- (D) 16
- (E) 18

Q8: A customer bought an article for Rs. 408 after getting two successive discounts on its marked price of Rs. 640. If the first discount was 25%, find the second discount given by the shopkeeper.

एक ग्राहक ने Rs. 640 अंकित मूल्य वाली वस्तु को दो क्रमागत छूट मिलने के बाद Rs. 408 में खरीदा। यदि पहली छूट 25% थी, तो दुकानदार द्वारा दी गई दूसरी छूट ज्ञात कीजिए।

- (A) 10%
- (B) 12%
- (C) 20%
- (D) 25%
- (E) 15%

Q9: A man invests equal amounts at the same rate of simple interest for X years and (X + 6) years. If the ratio of interest received by him is 2 : 5 respectively, find the value of X + 6.

एक व्यक्ति समान राशि को समान साधारण ब्याज दर पर x वर्षों और (x + 6) वर्षों के लिए

निवेश करता है। यदि उसे प्राप्त ब्याज का अनुपात क्रमशः 2 : 5 है, तो $X + 6$ का मान ज्ञात कीजिए।

- (A) 8
- (B) 10
- (C) 12
- (D) 14
- (E) 6

Q10: A mixture of milk and water contains 9 liters of water. When 6 liters of milk and 10 liters of water are added to the mixture, the concentration of milk becomes 75%. Find the total quantity of the initial mixture.

दूध और पानी के मिश्रण में 9 लीटर पानी है। जब मिश्रण में 6 लीटर दूध और 10 लीटर पानी मिलाया जाता है, तो मिश्रण में दूध की सांद्रता 75% हो जाती है। प्रारंभिक मिश्रण की कुल मात्रा ज्ञात कीजिए।

- (A) 48 liters
- (B) 54 liters
- (C) 66 liters
- (D) 60 liters
- (E) 72 liters

Q11: A mixture of milk and water initially contains 12 liters of water. When 6 liters of milk and 18 liters of water are added to it, the concentration of milk in the mixture becomes 75%. Find the total quantity of the initial mixture.

दूध और पानी के एक मिश्रण में प्रारंभ में 12 लीटर पानी है। जब इसमें 6 लीटर दूध और 18 लीटर पानी मिलाया जाता है, तो मिश्रण में दूध की सांद्रता 75% हो जाती है। प्रारंभिक मिश्रण की कुल मात्रा ज्ञात कीजिए।

- (A) 84 liters
- (B) 88 liters
- (C) 92 liters
- (D) 96 liters
- (E) 104 liters

Q12: Pipe P alone can fill a tank in 10 minutes. Two outlet pipes Q and R can empty the same tank in 30 minutes and 60 minutes respectively. If all three pipes are opened together and after X minutes pipe Q is closed, the remaining tank is filled by pipes P and R together in 9 minutes. Find X.

पाइप P अकेले एक टंकी को 10 मिनट में भर सकता है। दो निकासी पाइप Q और R उसी

टंकी को क्रमशः 30 मिनट और 60 मिनट में खाली कर सकते हैं। यदि तीनों पाइप एक साथ खोले जाते हैं और x मिनट बाद पाइप Q बंद कर दिया जाता है, तो शेष टंकी पाइप P और R द्वारा मिलकर 9 मिनट में भर जाती है। x ज्ञात कीजिए।

- (A) 5
- (B) 8
- (C) 10
- (D) 12
- (E) 15

Q13: Two travelers are initially $(D + 25)$ km apart and start moving towards each other at the same time. After 20 minutes, the distance between them becomes $(2D - 70)$ km. If their speeds are 45 km/hr and 30 km/hr, find the difference between the distances traveled by them when they meet.

दो यात्री प्रारंभ में $(D + 25)$ किमी दूर हैं और एक ही समय पर एक-दूसरे की ओर चलना शुरू करते हैं। 20 मिनट बाद उनके बीच की दूरी $(2D - 70)$ किमी हो जाती है। यदि उनकी चाल 45 किमी/घंटा और 30 किमी/घंटा है, तो मिलने तक दोनों द्वारा तय की गई दूरियों का अंतर ज्ञात कीजिए।

- (A) 15 km
- (B) 21 km
- (C) 24 km
- (D) 18 km
- (E) 19 km

Q14: When the upstream speed of a boat is reduced by 20%, the resulting speed becomes equal to the speed of the current. If the speed of the boat in still water is 90 km/hr, find the downstream speed of the boat.

जब एक नाव की धारा के विपरीत चाल में 20% की कमी की जाती है, तो प्राप्त चाल धारा की चाल के बराबर हो जाती है। यदि स्थिर जल में नाव की चाल 90 किमी/घंटा है, तो नाव की धारा के अनुकूल चाल ज्ञात कीजिए।

- (A) 120 km/hr
- (B) 130 km/hr
- (C) 140 km/hr
- (D) 150 km/hr
- (E) 160 km/hr

Q15: A solid sphere of radius 15 cm is melted and recast into a cylinder whose radius is the same as that of the sphere. Find the height of the cylinder.

15 सेमी त्रिज्या वाले एक ठोस गोले को पिघलाकर एक बेलन में ढाला जाता है, जिसकी त्रिज्या गोले की त्रिज्या के बराबर है। बेलन की ऊँचाई ज्ञात कीजिए।

- (A) 15 cm
- (B) 18 cm
- (C) 20 cm
- (D) 24 cm
- (E) 30 cm

Q16: The ratio of A to B is 5:3. If the average of both numbers is 48 less than A, find 25% of B.

A और B का अनुपात 5:3 है। यदि दोनों संख्याओं का औसत A से 48 कम है, तो B का 25% ज्ञात कीजिए।

- (A) 30
- (B) 48
- (C) 42
- (D) 36
- (E) 60

Q17: In a colony, 35% of the residents are male and the rest are female. If 30% of the females are graduates and the number of non-graduate females is 910, find the number of males in the colony.

एक कॉलोनी में 35% निवासी पुरुष हैं और शेष महिलाएं हैं। यदि महिलाओं में से 30% स्नातक हैं और गैर-स्नातक महिलाओं की संख्या 910 है, तो कॉलोनी में पुरुषों की संख्या ज्ञात कीजिए।

- (A) 650
- (B) 900
- (C) 700
- (D) 800
- (E) 1000

Q18: The average of 61 numbers is 54. If the average of the first 30 numbers is 50 and the average of the last 30 numbers is 56, find the middle number.

61 संख्याओं का औसत 54 है। यदि पहली 30 संख्याओं का औसत 50 और अंतिम 30 संख्याओं का औसत 56 है, तो बीच वाली संख्या ज्ञात कीजिए।

- (A) 108
- (B) 114
- (C) 120

- (D) 96
(E) 102

Q19: Two partners invested in a business in the ratio of 5:4. The first partner invested for 16 months and the second partner invested for 15 months. If $\frac{1}{5}$ of the total profit goes to charity and the first partner's share from the remaining profit is Rs 4800, find the second partner's profit share.

दो साझेदारों ने 5:4 के अनुपात में एक व्यवसाय में निवेश किया। पहले साझेदार ने 16 महीनों के लिए और दूसरे साझेदार ने 15 महीनों के लिए निवेश किया। यदि कुल लाभ का $\frac{1}{5}$ भाग दान में चला जाता है और शेष लाभ में पहले साझेदार का हिस्सा रु. 4800 है, तो दूसरे साझेदार का लाभांश ज्ञात कीजिए।

- (A) 4200
(B) 3000
(C) 4800
(D) 3600
(E) 2400

Q20: The population of a city was 40,000. It increased by 20% in the first year and decreased by 15% in the second year. Find the population at the end of two years.

एक शहर की जनसंख्या 40,000 थी। पहले वर्ष में यह 20% बढ़ी और दूसरे वर्ष में 15% घट गई। दो वर्षों के अंत में जनसंख्या ज्ञात कीजिए।

- (A) 41600
(B) 38000
(C) 43200
(D) 40000
(E) 40800

Answers:

- | | |
|------|------|
| 1. D | 7. A |
| 2. B | 8. E |
| 3. E | 9. B |
| 4. B | 10.D |
| 5. D | 11.D |
| 6. C | 12.A |

13.E
14.B
15.C
16.D

17.C
18.B
19.D
20.E

Solutions:

1. Option D

Let the number of male employees in each company be M.

Let the number of female employees in company P be F.

Female employees in company Q are 40% less than company P.

So, female employees in company Q = 60% of F = $60F/100 = 3F/5$

Initial total employees in company P = M + F

Initial total employees in company Q = M + $3F/5$

After change in company P:

Male employees in company P decrease by 40%.

New male employees in company P = $60M/100 = 3M/5$

Female employees in company P increase by 20%.

New female employees in company P = $120F/100 = 6F/5$

Now total employees in company P become equal to total employees in company Q.

So:

$$3M/5 + 6F/5 = M + 3F/5$$

$$6F/5 - 3F/5 = M - 3M/5$$

$$3F/5 = 2M/5$$

$$3F = 2M$$

$$M/F = 3/2$$

Let M = 3x and F = 2x.

Initial total employees in company P = M + F = 3x + 2x = 5x

Initial total employees in company Q = M + $3F/5$

Initial total employees in company Q = 3x + $3 \times 2x/5$

Initial total employees in company Q = 3x + $6x/5$

Initial total employees in company Q = $15x/5 + 6x/5 = 21x/5$

Required ratio = 5x : $21x/5$

Required ratio = 25x : 21x

Required ratio = 25:21

2. Option B

Average tea consumption per working day = 210 cups

Number of working days in a year = 264

Total tea consumption in a year = 210×264

Total tea consumption in a year = 55440 cups

Average monthly tea consumption per employee = 35 cups

Average yearly tea consumption per employee = 35×12

Average yearly tea consumption per employee = 420 cups

Number of employees = Total yearly tea consumption / Yearly consumption per employee

Number of employees = $55440/420$

Number of employees = 132

3. Option E

Distance between city A and city B = 540 km

Speed of van = X km/hr

Speed of car = $(X + 30)$ km/hr

Time taken by van = $540/X$ hours

Time taken by car = $540/(X + 30)$ hours

Car takes 3 hours less than van.

So:

$$540/X - 540/(X + 30) = 3$$

$$540[(X + 30) - X] / [X(X + 30)] = 3$$

$$540 \times 30 / [X(X + 30)] = 3$$

$$16200 = 3X(X + 30)$$

$$5400 = X(X + 30)$$

$$X^2 + 30X - 5400 = 0$$

$$X^2 + 90X - 60X - 5400 = 0$$

$$X(X + 90) - 60(X + 90) = 0$$

$$(X + 90)(X - 60) = 0$$

$$X = 60$$

Required value = $X + 8$

Required value = $60 + 8 = 68$

4. Option B

Five consecutive multiples of 10 have an average of 70.

So, the five numbers are:

50, 60, 70, 80, 90

Highest number = 90

Second smallest number = 60

Difference = $90 - 60 = 30$

This difference is equal to the largest even number in a set of four consecutive even numbers.

So, largest even number = 30

The four consecutive even numbers are:

24, 26, 28, 30

Smallest even number = 24

Square of the smallest even number = 24×24

Square = 576

5. Option D

Perimeter of the right-angled triangle = 84 meters

Hypotenuse = 35 meters

Sum of the other two sides = $84 - 35$

Sum of the other two sides = 49 meters

Let the two smaller sides be x and $49 - x$.

According to Pythagoras theorem:

$$x^2 + (49 - x)^2 = 35^2$$

$$x^2 + 2401 - 98x + x^2 = 1225$$

$$2x^2 - 98x + 2401 = 1225$$

$$2x^2 - 98x + 1176 = 0$$

$$x^2 - 49x + 588 = 0$$

$$x^2 - 21x - 28x + 588 = 0$$

$$x(x - 21) - 28(x - 21) = 0$$

$$(x - 21)(x - 28) = 0$$

So, the two sides are 21 meters and 28 meters.

The base is the smallest side.

Therefore, the base is 21 meters.

6. Option C

Average salary of Kiran and Megha = Rs. 4500.

Total initial salary = $4500 \times 2 = \text{Rs. } 9000$.

Let Kiran's initial salary be Rs. x .

Then Megha's initial salary = Rs. $(9000 - x)$.

Total salary increases by 30%.

Total increase = 30% of 9000.

Total increase = $9000 \times 30/100 = \text{Rs. } 2700$.

Increase in Kiran's salary = 40% of $x = 40x/100$.

Increase in Megha's salary = 25% of $(9000 - x) = 25(9000 - x)/100$.

So,

$$40x/100 + 25(9000 - x)/100 = 2700$$

$$40x + 25(9000 - x) = 270000$$

$$40x + 225000 - 25x = 270000$$

$$15x = 270000 - 225000$$

$$15x = 45000$$

$$x = 45000/15$$

$$x = 3000$$

Kiran's initial salary = Rs. 3000.

Megha's initial salary = $9000 - 3000 = \text{Rs. } 6000$.

Difference = $6000 - 3000 = \text{Rs. } 3000$.

Required percentage = $3000/3000 \times 100 = 100\%$.

7. Option A

Let the initial number of people in the group be n .

Initial average weight = 45 kg.

Initial total weight = $45n$ kg.

One person weighing 63 kg leaves the group.

New number of people = $n - 1$.

New average weight = $45 - 2 = 43$ kg.

New total weight = $43(n - 1)$.

So,

$$45n - 63 = 43(n - 1)$$

$$45n - 63 = 43n - 43$$

$$45n - 43n = 63 - 43$$

$$2n = 20$$

$$n = 20/2$$

$$n = 10$$

8. Option E

Marked price of the article = Rs. 640.

First discount = 25%.

Price after first discount = $640 \times 75/100$.

Price after first discount = Rs. 480.

Final selling price after second discount = Rs. 408.

Second discount amount = $480 - 408 = \text{Rs. } 72$.

Second discount percentage = $72/480 \times 100$.

Second discount percentage = 15%.

9. Option B

For simple interest, when principal and rate are the same, interest is directly proportional to time.

Time periods are X years and $(X + 6)$ years.

Ratio of interest = $2 : 5$.

So,

$$X/(X + 6) = 2/5$$

$$5X = 2(X + 6)$$

$$5X = 2X + 12$$

$$5X - 2X = 12$$

$$3X = 12$$

$$X = 12/3$$

$$X = 4$$

Required value = $X + 6$.

$$X + 6 = 4 + 6 = 10.$$

10. Option D

Let the initial quantity of the mixture be x liters.

Initial water = 9 liters.

Initial milk = $x - 9$ liters.

Now, 6 liters of milk and 10 liters of water are added.

Final milk = $x - 9 + 6 = x - 3$ liters.

Final total mixture = $x + 6 + 10 = x + 16$ liters.

Milk concentration in the final mixture = 75%.

So,

$$(x - 3)/(x + 16) = 75/100$$

$$(x - 3)/(x + 16) = 3/4$$

$$4(x - 3) = 3(x + 16)$$

$$4x - 12 = 3x + 48$$

$$4x - 3x = 48 + 12$$

$$x = 60$$

11. Option D

Let the initial quantity of milk be x liters.

Initial quantity of water = 12 liters

Initial total mixture = $x + 12$ liters

Milk added = 6 liters

Water added = 18 liters

Final quantity of milk = $x + 6$

Final total mixture = $x + 12 + 6 + 18 = x + 36$

Final concentration of milk = $75\% = 75/100 = 3/4$

So, $(x + 6)/(x + 36) = 3/4$

Cross multiply:

$$4(x + 6) = 3(x + 36)$$

$$4x + 24 = 3x + 108$$

$$4x - 3x = 108 - 24$$

$$x = 84$$

Initial total mixture = $84 + 12 = 96$ liters

12. Option A

Pipe P fills the tank in 10 minutes.

P's 1-minute work = $1/10$

Pipe Q empties the tank in 30 minutes.

Q's 1-minute work = $-1/30$

Pipe R empties the tank in 60 minutes.

R's 1-minute work = $-1/60$

When all three pipes are opened together:

Net 1-minute work = $1/10 - 1/30 - 1/60$

LCM of 10, 30 and 60 = 60

Net 1-minute work = $6/60 - 2/60 - 1/60 = 3/60 = 1/20$

So, work done in X minutes = $X/20$

After X minutes, pipe Q is closed.

Now, pipes P and R work together.

Net 1-minute work of P and R = $1/10 - 1/60$

Net work = $6/60 - 1/60 = 5/60 = 1/12$

They fill the remaining tank in 9 minutes.

Work done by P and R in 9 minutes = $9 \times \frac{1}{12} = \frac{9}{12} = \frac{3}{4}$

So, work done by all three pipes before Q is closed = $1 - \frac{3}{4} = \frac{1}{4}$

Therefore, $\frac{X}{20} = \frac{1}{4}$

$X = \frac{20}{4} = 5$

13. Option E

Initial distance between the two travelers = $D + 25$ km

Distance after 20 minutes = $2D - 70$ km

Speed of first traveler = 45 km/hr

Speed of second traveler = 30 km/hr

Relative speed = $45 + 30 = 75$ km/hr

Time = 20 minutes = $\frac{20}{60} = \frac{1}{3}$ hour

Distance covered together in 20 minutes = $75 \times \frac{1}{3} = 25$ km

So, decrease in distance = 25 km

Initial distance – Distance after 20 minutes = 25

$(D + 25) - (2D - 70) = 25$

$D + 25 - 2D + 70 = 25$

$95 - D = 25$

$D = 70$

Initial distance = $D + 25 = 70 + 25 = 95$ km

Time taken to meet = Initial distance/Relative speed

Time = $\frac{95}{75} = \frac{19}{15}$ hours

Difference in their speeds = $45 - 30 = 15$ km/hr

Difference in distance traveled by them = $15 \times \frac{19}{15} = 19$ km

14. Option B

Let the upstream speed of the boat be U km/hr.

When upstream speed is reduced by 20%, it becomes 80% of U .

So, speed of current = 80% of $U = \frac{80U}{100} = \frac{4U}{5}$

Speed of boat in still water = Upstream speed + Speed of current

$90 = U + \frac{4U}{5}$

$90 = \frac{9U}{5}$

$U = 90 \times \frac{5}{9}$

$U = 50$ km/hr

Speed of current = $\frac{4U}{5} = 4 \times \frac{50}{5} = 40$ km/hr

Downstream speed = Speed in still water + Speed of current

Downstream speed = $90 + 40 = 130$ km/hr

15. Option C

When a sphere is melted and recast into a cylinder, their volumes are equal.

Radius of sphere = 15 cm

Radius of cylinder = 15 cm

Volume of sphere = $\frac{4}{3} \times \pi \times r \times r \times r$

Volume of cylinder = $\pi \times r \times r \times h$

Since both volumes are equal:

$\frac{4}{3} \times \pi \times 15 \times 15 \times 15 = \pi \times 15 \times 15 \times h$

Cancel $\pi \times 15 \times 15$ from both sides.

$h = \frac{4}{3} \times 15$

$h = \frac{60}{3} = 20$ cm

16. Option D

Ratio of A to B = 5:3.

Let A = 5x and B = 3x.

Average of A and B = $(5x + 3x)/2$

Average = $8x/2 = 4x$

Given, average is 48 less than A.

So, A - average = 48

$5x - 4x = 48$

$x = 48$

$B = 3x = 3 \times 48 = 144$

Required value = 25% of B

25% of 144 = $25/100 \times 144$

25% of 144 = 36

17. Option C

In the colony, males = 35%.

So, females = $100\% - 35\% = 65\%$.

30% of females are graduates.

So, non-graduate females = $100\% - 30\% = 70\%$ of females.

Given, non-graduate females = 910.

So, 70% of females = 910

Females = $910 \times 100/70$

Females = 1300

Now, 65% of total residents = 1300.

Total residents = $1300 \times 100/65$

Total residents = 2000

Males = 35% of 2000

Males = $35/100 \times 2000$

Males = 700

18. Option B

Average of 61 numbers = 54.

Total of 61 numbers = $61 \times 54 = 3294$.

Average of first 30 numbers = 50.

Sum of first 30 numbers = $30 \times 50 = 1500$.

Average of last 30 numbers = 56.

Sum of last 30 numbers = $30 \times 56 = 1680$.

Sum of first 30 and last 30 numbers = $1500 + 1680 = 3180$.

Middle number = Total of 61 numbers - Sum of first 30 and last 30 numbers

Middle number = $3294 - 3180$

Middle number = 114

19. Option D

Profit is divided in the ratio of investment $\times$ time.

Investment ratio of first partner and second partner = 5:4.

Time ratio = 16:15.

Profit ratio = $5 \times 16 : 4 \times 15$

Profit ratio = 80:60

Profit ratio = 4:3

The charity amount does not change the ratio of the remaining profit shared between the partners.

First partner's share = 4 parts = Rs 4800.

So, 1 part = $4800/4 = \text{Rs } 1200$.

Second partner's share = 3 parts.

Second partner's share = $3 \times 1200 = \text{Rs } 3600$.

20. Option E

Initial population = 40,000.

In the first year, population increased by 20%.

Population after first year = $40000 \times 120/100$

Population after first year = 48000

In the second year, population decreased by 15%.

Population after second year = $48000 \times 85/100$

Population after second year = 40800

CHECKLIST

BY

AASHISH

ARORA

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $12x^2 - 23x + 5 = 0$

II. $6y^2 + 41y + 65 = 0$

2.) I. $2x^2 - 10\sqrt{5}x + 60 = 0$

II. $3y^2 - 9\sqrt{5}y + 30 = 0$

3.) I. $8x^2 + 17x + 9 = 0$

II. $6y^2 - y - 5 = 0$

4.) I. $3x^2 - 28x + 44 = 0$

II. $8y^2 - 19y + 6 = 0$

5.) I. $x^2 + 7x + 10 = 0$

II. $5y^2 - 6y - 32 = 0$

6.) I. $3x^2 - 9\sqrt{3}x + 18 = 0$

II. $2y^2 - 14\sqrt{3}y + 72 = 0$

7.) I. $6x^2 + 19x - 130 = 0$

II. $8y^2 - 18y - 143 = 0$

8.) I. $12x^2 - 61x + 70 = 0$

II. $3y^2 - y - 4 = 0$

9.) I. $7x^2 + 61x + 84 = 0$

II. $5y^2 - 56y + 128 = 0$

10.) I. $10x^2 + 49x + 60 = 0$

II. $2y^2 + 17y + 30 = 0$

11.) I. $2x^2 - 14\sqrt{2}x + 48 = 0$

II. $3y^2 - 9\sqrt{2}y + 12 = 0$

12.) I. $5x^2 + 33x + 54 = 0$

II. $3y^2 - 10y - 57 = 0$

13.) I. $5x^2 - 14x + 8 = 0$

II. $4y^2 - 23y - 35 = 0$

14.) I. $3x^2 - 14x - 24 = 0$

II. $5y^2 + 21y + 22 = 0$

15.) I. $x^2 - 5\sqrt{6}x + 24 = 0$

II. $2y^2 - 6\sqrt{6}y + 24 = 0$

16.) I. $8x^2 + 6x + 1 = 0$

II. $5y^2 - 51y + 112 = 0$

17.) I. $4x^2 - 16x - 9 = 0$

II. $10y^2 + 23y + 9 = 0$

18.) I. $5x^2 + 42x + 88 = 0$

II. $7y^2 + 47y + 76 = 0$

19.) I. $3x^2 - 9\sqrt{7}x + 42 = 0$

II. $2y^2 - 10\sqrt{7}y + 84 = 0$

20.) I. $2x^2 + 7x + 6 = 0$

II. $12y^2 + 41y + 22 = 0$

Answers:

1. A

2. D

3. B

4. D

5. E

6. B

7. C

8. A

9. B

10. D

11. A

12. E

13. C

14. A

15. C

16. B

17. D

18. E

19. E

20. C

Solutions:

$$(1) x = \frac{5}{3}, \frac{1}{4}$$

$$y = \frac{5}{2}, -\frac{13}{3}$$

$$(2) x = 2\sqrt{5}, 3\sqrt{5}$$

$$y = \sqrt{5}, 2\sqrt{5}$$

$$(3) x = -1, -\frac{9}{8}$$

$$y = 1, -\frac{5}{6}$$

$$(4) x = \frac{22}{3}, 2$$

$$y = 2, \frac{3}{8}$$

$$(5) x = -2, -5$$

$$y = \frac{16}{5}, -2$$

$$(6) x = \sqrt{3}, 2\sqrt{3}$$

$$y = 3\sqrt{3}, 4\sqrt{3}$$

$$(7) x = \frac{10}{3}, -\frac{13}{2}$$

$$y = \frac{11}{2}, -\frac{13}{4}$$

$$(8) x = \frac{10}{3}, \frac{7}{4}$$

$$y = \frac{4}{3}, -1$$

$$(9) x = -\frac{12}{7}, -7$$

$$y = 8, \frac{16}{5}$$

$$(10) x = -\frac{12}{5}, -\frac{5}{2}$$

$$y = -\frac{5}{2}, -6$$

$$(11) x = 3\sqrt{2}, 4\sqrt{2}$$

$$y = \sqrt{2}, 2\sqrt{2}$$

$$(12) x = -3, -\frac{18}{5}$$

$$y = \frac{19}{3}, -3$$

$$(13) x = 2, \frac{4}{5}$$

$$y = 7, -\frac{5}{4}$$

$$(14) x = 6, -\frac{4}{3}$$

$$y = -2, -\frac{11}{5}$$

$$(15) x = \sqrt{6}, 4\sqrt{6}$$

$$y = \sqrt{6}, 2\sqrt{6}$$

$$(16) x = -\frac{1}{4}, -\frac{1}{2}$$

$$y = 7, \frac{16}{5}$$

$$(17) x = \frac{9}{2}, -\frac{1}{2}$$

$$y = -\frac{1}{2}, -\frac{9}{5}$$

$$(18) x = -4, -\frac{22}{5}$$

$$y = -\frac{19}{7}, -4$$

$$(19) x = \sqrt{7}, 2\sqrt{7}$$

$$y = 2\sqrt{7}, 3\sqrt{7}$$

$$(20) x = -\frac{3}{2}, -2$$

$$y = -\frac{2}{3}, -\frac{11}{4}$$

4. WRONG NUMBER SERIES

(1) 1840, 1864, 1899, 1937, 2002, 2070

- (a) 2002
- (b) 1937
- (c) 2070
- (d) 1864
- (e) None of these

(2) 2450, 2432, 2401, 2357, 2306, 2230

- (a) 2432
- (b) 2230
- (c) 2450
- (d) 2306
- (e) None of these

(3) 126, 276, 836, 3408, 17052, 102320

- (a) 836
- (b) 17052
- (c) 276
- (d) 3408
- (e) None of these

(4) 93004, 46495, 15488, 3865, 768, 125

- (a) 15488
- (b) 3865
- (c) 768
- (d) 93004
- (e) None of these

(5) 415, 432, 852, 1294, 2169, 3475

- (a) 852
- (b) 415
- (c) 2169
- (d) 1294
- (e) None of these

(6) 512, 1024, 2031, 4053, 8095, 16177

- (a) 1024
- (b) 2031
- (c) 8095
- (d) 512
- (e) None of these

(7) 880, 895, 925, 995, 1105, 1345

(a) 895

(b) 1345

(c) 1105

(d) 995

(e) None of these

(8) 140, 414, 222, 642, 351, 1017

(a) 351

(b) 414

(c) 222

(d) 1017

(e) None of these

(9) 724, 740, 777, 856, 1019, 1357

(a) 1019

(b) 740

(c) 777

(d) 724

(e) None of these

(10) 3000, 2979, 2935, 2865, 2718, 2427

(a) 2718

(b) 3000

(c) 2935

(d) 2427

(e) None of these

(11) 92, 203, 582, 2343, 11700, 70215

(a) 203

(b) 11700

(c) 582

(d) 2343

(e) None of these

(12) 36540, 18288, 6120, 1572, 348, 100

(a) 348

(b) 36540

(c) 100

(d) 1572

(e) None of these

(13) 580, 590, 620, 688, 830, 1040

(a) 688

(b) 830

(c) 1040

(d) 590

(e) None of these

(14) 720, 733, 707, 759, 655, 873

(a) 707

(b) 720

(c) 759

(d) 655

(e) None of these

(15) 84, 492, 2455, 9762, 29265,
58506

(a) 492

(b) 58506

(c) 2455

(d) 29265

(e) None of these

(16) 250.5, 262.75, 272.25, 301.25,
359.25, 475.25

(a) 262.75

(b) 272.25

(c) 359.25

(d) 475.25

(e) None of these

(17) 1980, 1952, 1907, 1831, 1750,
1630

(a) 1907

(b) 1630

(c) 1980

(d) 1831

(e) None of these

(18) 395, 421, 465, 565, 769, 1181

(a) 465

(b) 421

(c) 769

(d) 1181

(e) None of these

(19) 40, 121, 365, 1089, 3305, 9931

(a) 121

(b) 40

(c) 3305

(d) 9931

(e) None of these

(20) 1020, 1034, 1064, 1088, 1132,
1190

(a) 1132

(b) 1190

(c) 1064

(d) 1034

(e) None of these

Answers:

(1) b

(2) d

(3) a

(4) e

(5) c

(6) a

(7) d

(8) b

(9) e

(10) c

(11) a

(12) d

(13) b

(14) e

(15) c

(16) a

(17) d

(18) b

(19) e

(20) c

Solutions:

(1) +24, +35, +46, +57, +68; second differences: +11, +11, +11, +11.

(2) -18, -31, -44, -57, -70; second differences: -13, -13, -13, -13.

(3) $\times 2 + 24$, $\times 3 + 20$, $\times 4 + 16$, $\times 5 + 12$, $\times 6 + 8$.

(4) $\div 2 - 11$, $\div 3 - 9$, $\div 4 - 7$, $\div 5 - 5$, $\div 6 - 3$.

(5) Previous two terms +5, +10, +15, +20.

(6) $\times 2 - 5$, $\times 2 - 7$, $\times 2 - 9$, $\times 2 - 11$, $\times 2 - 13$.

(7) +15, +30, +60, +120, +240; each next difference is $\times 2$.

(8) $\times 3 - 12$, $\div 2 + 18$, $\times 3 - 24$, $\div 2 + 30$, $\times 3 - 36$.

(9) +16, +37, +79, +163, +331; second differences: +21, +42, +84, +168.

(10) -21, -39, -75, -147, -291; second differences: -18, -36, -72, -144.

(11) $\times 2 + 15$, $\times 3 - 15$, $\times 4 + 15$, $\times 5 - 15$, $\times 6 + 15$.

(12) $\div 2 + 18$, $\div 3 + 24$, $\div 4 + 30$, $\div 5 + 36$, $\div 6 + 42$.

(13) $+(2^3 + 2)$, $+(3^3 + 3)$, $+(4^3 + 4)$, $+(5^3 + 5)$, $+(6^3 + 6)$.

(14) +13, -26, +52, -104, +208.

(15) $\times 6 - 12$, $\times 5 - 15$, $\times 4 - 18$, $\times 3 - 21$, $\times 2 - 24$.

(16) +7.25, +14.5, +29, +58, +116; each next difference is $\times 2$.

(17) $-(4 \times 7)$, $-(5 \times 9)$, $-(6 \times 11)$, $-(7 \times 13)$, $-(8 \times 15)$.

(18) +22, +48, +100, +204, +412; second differences: +26, +52, +104, +208.

(19) $\times 3 + 1$, $\times 3 + 2$, $\times 3 + 4$, $\times 3 + 8$, $\times 3 + 16$.

(20) +14, +22, +32, +44, +58; second differences: +8, +10, +12, +14.

5. MISSING NUMBER SERIES

(1) 36, 47, 71, ?, 166, 241

- (a) 108
- (b) 110
- (c) 112
- (d) 114
- (e) None of these

(2) 7, 15, ?, 205, 1041, 6271

- (a) 49
- (b) 47
- (c) 51
- (d) 53
- (e) None of these

(3) 4548, 2286, ?, 210, 60, 30

- (a) 772
- (b) 774
- (c) 778
- (d) 776
- (e) None of these

(4) 84, 93, 182, 285, ?, 787

- (a) 480
- (b) 482
- (c) 484
- (d) 486
- (e) None of these

(5) 512, 495, 466, ?, 364, 287

- (a) 423
- (b) 421
- (c) 425

(d) 427

(e) None of these

(6) 15.5, 18, 23.5, ?, 47.5, 68

- (a) 30
- (b) 31
- (c) 32
- (d) 33

(e) None of these

(7) 4600, 4582, 4540, ?, 4360, 4210

- (a) 4464
- (b) 4466
- (c) 4468
- (d) 4470

(e) None of these

(8) 5, 8, 20, ?, 125, 260

- (a) 51
- (b) 53
- (c) 55
- (d) 57

(e) None of these

(9) 123, 253, 520, ?, 2150, 4335

- (a) 1057
- (b) 1059
- (c) 1063
- (d) 1061

(e) None of these

(10) 72, 85, 112, ?, 224, 317

- (a) 157
- (b) 155
- (c) 159
- (d) 161
- (e) None of these

(11) 3600, 900, 2700, ?, 2160, 360

- (a) 520
- (b) 530
- (c) 540
- (d) 550
- (e) None of these

(12) 28, 61, 129, ?, 551, 1119

- (a) 267
- (b) 269
- (c) 271
- (d) 273
- (e) None of these

(13) 810, 797, 773, ?, 692, 635

- (a) 732
- (b) 734
- (c) 736
- (d) 738
- (e) None of these

(14) 9, 23, 75, ?, 1543, 9267

- (a) 303
- (b) 305
- (c) 307
- (d) 309
- (e) None of these

(15) 2400, 2412, 2448, ?, 2640, 2820

- (a) 2520
- (b) 2510
- (c) 2530
- (d) 2540
- (e) None of these

(16) 66, 83, 113, ?, 220, 301

- (a) 154
- (b) 156
- (c) 158
- (d) 160
- (e) None of these

(17) 145, 161, 193, ?, 305, 385

- (a) 239
- (b) 241
- (c) 243
- (d) 245
- (e) None of these

(18) 3148, 1581, 536, ?, 42, 22

- (a) 139
- (b) 141
- (c) 143
- (d) 145
- (e) None of these

(19) 18, 54, 132, ?, 624, 1290

- (a) 294
- (b) 292
- (c) 296
- (d) 298
- (e) None of these

(20) 625, 630, 655, ?, 1405, 4530

- (a) 775
- (b) 780
- (c) 785
- (d) 790
- (e) None of these

Answers:

- (1) b
- (2) a

- (3) d
 (4) b
 (5) a
 (6) d
 (7) c
 (8) b
 (9) d
 (10) a
 (11) c
 (12) b
 (13) d
 (14) c
 (15) a
 (16) c
 (17) b
 (18) d
 (19) a
 (20) b

Solutions:

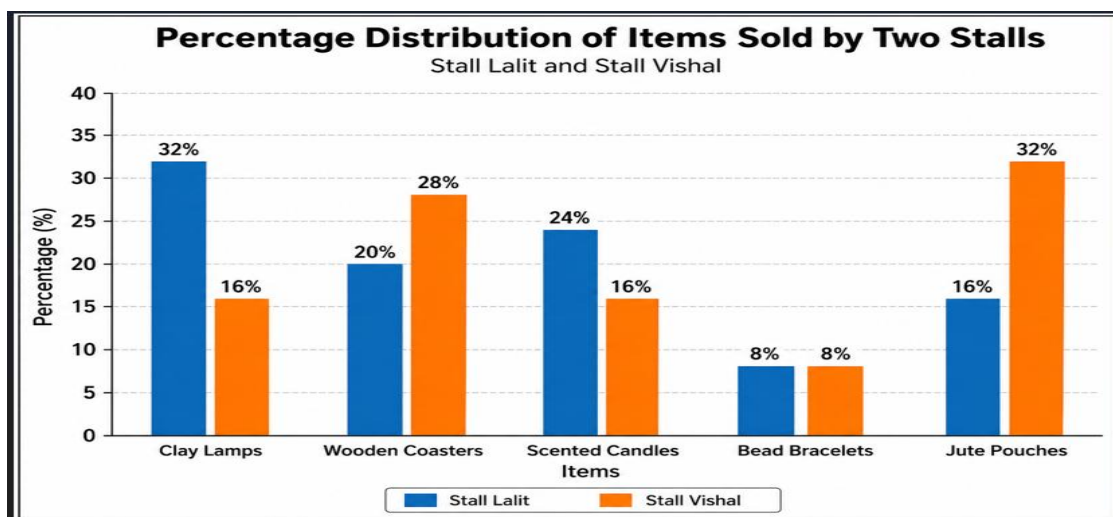
- (1) +11, +24, +39, +56, +75
 Second difference: +13, +15, +17, +19
 (2) $\times 2 + 1^2$, $\times 3 + 2^2$, $\times 4 + 3^2$, $\times 5 + 4^2$,
 $\times 6 + 5^2$
 (3) $\div 2 + 12$, $\div 3 + 14$, $\div 4 + 16$, $\div 5 + 18$,
 $\div 6 + 20$
 (4) Sum of previous two numbers +5,
 +10, +15, +20
 (5) -17, -29, -43, -59, -77
 Second difference: -12, -14, -16, -18

- (6) +2.5, +5.5, +9.5, +14.5, +20.5
 (7) -18, -42, -72, -108, -150
 (8) $+1^3 + 2$, $+2^3 + 4$, $+3^3 + 6$, $+4^3 + 8$,
 $+5^3 + 10$
 (9) $\times 2 + 7$, $\times 2 + 14$, $\times 2 + 21$, $\times 2 + 28$,
 $\times 2 + 35$
 (10) +13, +27, +45, +67, +93
 Second difference: +14, +18, +22, +26
 (11) $\div 4$, $\times 3$, $\div 5$, $\times 4$, $\div 6$
 (12) $\times 2 + 5$, $\times 2 + 7$, $\times 2 + 11$, $\times 2 + 13$,
 $\times 2 + 17$
 (13) -13, -24, -35, -46, -57
 (14) $\times 2 + 5$, $\times 3 + 6$, $\times 4 + 7$, $\times 5 + 8$, $\times 6 +$
 9
 (15) +12, +36, +72, +120, +180
 (16) +17, +30, +45, +62, +81
 Second difference: +13, +15, +17, +19
 (17) +16, +32, +48, +64, +80
 (18) $\div 2 + 7$, $\div 3 + 9$, $\div 4 + 11$, $\div 5 + 13$,
 $\div 6 + 15$
 (19) $\times 2 + 18$, $\times 2 + 24$, $\times 2 + 30$, $\times 2 + 36$,
 $\times 2 + 42$
 (20) $+5^1$, $+5^2$, $+5^3$, $+5^4$, $+5^5$

6. DATA INTERPRETATION

Set 1: The bar graph shows the percentage distribution of five handmade gift items sold in two stalls, Lalit and Vishal. The direct percentages written in brackets should be used for solving the questions. Only these five items are sold in both stalls.

दंड आलेख दो स्टॉल, ललित और विशाल में बेची गई पाँच हस्तनिर्मित उपहार वस्तुओं का प्रतिशत वितरण दर्शाता है। प्रश्न हल करने के लिए कोष्ठक में लिखे सीधे प्रतिशत मानों का उपयोग करें। दोनों स्टॉलों में केवल ये पाँच वस्तुएँ ही बेची जाती हैं।



Q1: If the total number of gift items in Stall Vishal is 60% more than that in Stall Lalit, then the total number of Clay Lamps in both stalls together is what percentage of the total items in both stalls together? If the required percentage is $a\frac{b}{c}\%$, find $a + b + c$.

यदि स्टॉल विशाल में कुल उपहार वस्तुओं की संख्या स्टॉल ललित से 60% अधिक है, तो दोनों स्टॉलों में मिलाकर Clay Lamps की कुल संख्या, दोनों स्टॉलों में मिलाकर कुल वस्तुओं का कितना प्रतिशत है? यदि आवश्यक प्रतिशत $a\frac{b}{c}\%$ है, तो $a + b + c$ ज्ञात कीजिए।

- (A) 31
- (B) 33
- (C) 37
- (D) 41

(E) 43

Q2: If the total number of gift items in Stall Vishal is 80% more than that in Stall Lalit and the total number of Wooden Coasters in both stalls together is 1408, then find the difference between Clay Lamps in Stall Lalit and Bead Bracelets in Stall Vishal.

यदि स्टॉल विशाल में कुल उपहार वस्तुओं की संख्या स्टॉल ललित से 80% अधिक है और दोनों स्टॉलों में मिलाकर Wooden Coasters की कुल संख्या 1408 है, तो स्टॉल ललित में Clay Lamps और स्टॉल विशाल में Bead Bracelets के बीच अंतर ज्ञात कीजिए।

- (A) 352
- (B) 336
- (C) 368
- (D) 384
- (E) 320

Q3: If the total number of gift items in Stall Lalit and Stall Vishal is in the ratio 3:4 respectively, then Scented Candles in both stalls together are what percentage of the total items in both stalls together? If the required percentage is $a/b\%$, find $a + b + c$.

यदि स्टॉल ललित और स्टॉल विशाल में कुल उपहार वस्तुओं की संख्या का अनुपात क्रमशः 3:4 है, तो दोनों स्टॉलों में मिलाकर Scented Candles, दोनों स्टॉलों में मिलाकर कुल वस्तुओं का कितना प्रतिशत है? यदि आवश्यक प्रतिशत $a/b\%$ है, तो $a + b + c$ ज्ञात कीजिए।

- (A) 23
- (B) 25
- (C) 27
- (D) 31
- (E) 29

Q4: If the total number of gift items in Stall Vishal is 40% more than that in Stall Lalit and the total number of Bead Bracelets in both stalls together is 384, then find the total number of Clay Lamps in both stalls together.

यदि स्टॉल विशाल में कुल उपहार वस्तुओं की संख्या स्टॉल ललित से 40% अधिक है और दोनों स्टॉलों में मिलाकर Bead Bracelets की कुल संख्या 384 है, तो दोनों स्टॉलों में मिलाकर Clay Lamps की कुल संख्या ज्ञात कीजिए।

- (A) 1056
- (B) 1088
- (C) 1120

- (D) 1040
(E) 1152

Q5: If the number of Jute Pouches in Stall Lalit is 40% of the number of Jute Pouches in Stall Vishal, then the total number of gift items in Stall Vishal is what percent more than the total number of gift items in Stall Lalit?

यदि स्टॉल ललित में Jute Pouches की संख्या, स्टॉल विशाल में Jute Pouches की संख्या का 40% है, तो स्टॉल विशाल में कुल उपहार वस्तुएँ, स्टॉल ललित की कुल उपहार वस्तुओं से कितने प्रतिशत अधिक हैं?

- (A) 20%
(B) 15%
(C) 10%
(D) 25%
(E) 30%

Solutions:

1. Option C

Let total items in Lalit = $5x$

Total items in Vishal = $8x$

Clay Lamps in Lalit = $5x \times 32\% = 1.6x$

Clay Lamps in Vishal = $8x \times 16\% = 1.28x$

Total Clay Lamps = $1.6x + 1.28x = 2.88x$

Total items in both stalls = $5x + 8x = 13x$

Required percentage = $2.88x / 13x \times 100 = 288/13\% = 22 \frac{2}{13}\%$

$a + b + c = 22 + 2 + 13 = 37$

2. Option A

Let total items in Lalit = x

Total items in Vishal = 180% of $x = 9x/5$

Wooden Coasters in Lalit = $x \times 20\%$

Wooden Coasters in Vishal = $9x/5 \times 28\%$

Total Wooden Coasters = $0.2x + 0.504x = 0.704x$

$0.704x = 1408$

$x = 2000$

Total items in Vishal = $2000 \times \frac{9}{5} = 3600$

Clay Lamps in Lalit = $2000 \times 32\% = 640$

Bead Bracelets in Vishal = $3600 \times 8\% = 288$

Difference = $640 - 288 = 352$

3. Option E

Let total items in Lalit and Vishal be $3x$ and $4x$

Scented Candles in Lalit = $3x \times 24\% = 0.72x$

Scented Candles in Vishal = $4x \times 16\% = 0.64x$

Total Scented Candles = $1.36x$

Total items in both stalls = $7x$

Required percentage = $1.36x / 7x \times 100 = 136/7\% = 19\frac{3}{7}\%$

$a + b + c = 19 + 3 + 7 = 29$

4. Option B

Let total items in Lalit = x

Total items in Vishal = 140% of $x = 7x/5$

Bead Bracelets in Lalit = $x \times 8\%$

Bead Bracelets in Vishal = $7x/5 \times 8\%$

Total Bead Bracelets = $0.08x + 0.112x = 0.192x$

$0.192x = 384$

$x = 2000$

Total items in Vishal = $2000 \times \frac{7}{5} = 2800$

Clay Lamps in Lalit = $2000 \times 32\% = 640$

Clay Lamps in Vishal = $2800 \times 16\% = 448$

Total Clay Lamps = $640 + 448 = 1088$

5. Option D

Let total items in Lalit = L and total items in Vishal = V

Jute Pouches in Lalit = 16% of $L = 0.16L$

Jute Pouches in Vishal = 32% of $V = 0.32V$

Given, $0.16L = 40\%$ of $0.32V$

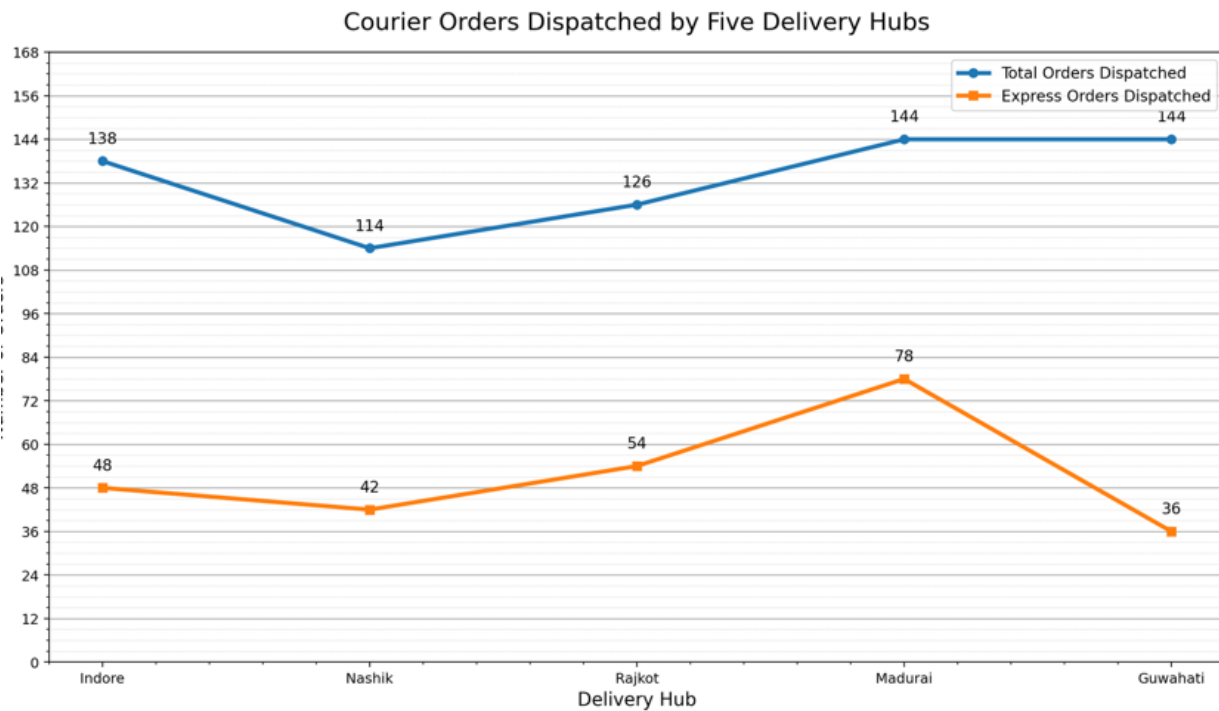
$0.16L = 0.128V$

$V/L = 0.16/0.128 = 1.25$

Stall Vishal has 25% more total items than Stall Lalit.

Set 2: The line graph shows the total number of courier orders dispatched and the number of express orders dispatched by five different delivery hubs: Indore, Nashik, Rajkot, Madurai and Guwahati. Total orders dispatched = Express orders dispatched + Standard orders dispatched. Study the line graph carefully and answer the following questions.

रेखा आलेख पांच अलग-अलग वितरण केंद्रों: इंदौर, नासिक, राजकोट, मदुरै और गुवाहाटी द्वारा भेजे गए कुल कूरियर ऑर्डरों और भेजे गए एक्सप्रेस ऑर्डरों की संख्या को दर्शाता है। भेजे गए कुल ऑर्डर = भेजे गए एक्सप्रेस ऑर्डर + भेजे गए स्टैंडर्ड ऑर्डर। रेखा आलेख का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Q1: The total number of standard orders dispatched by Guwahati is what percent more or less than the total number of standard orders dispatched by Nashik?

गुवाहाटी द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की कुल संख्या, नासिक द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की कुल संख्या से कितने प्रतिशत अधिक या कम है?

- (A) 33 1/3% more
- (B) 40% more
- (C) 25% less
- (D) 50% more
- (E) 60% more

Q2: Find the ratio of the total number of standard orders dispatched by Indore to the total number of standard orders dispatched by Rajkot.

इंदौर द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की कुल संख्या का, राजकोट द्वारा भेजे गए स्टैंडर्ड ऑर्डरों

की कुल संख्या से अनुपात ज्ञात कीजिए।

- (A) 5 : 4
- (B) 4 : 5
- (C) 6 : 5
- (D) 3 : 2
- (E) 5 : 6

Q3: Find the average number of standard orders dispatched by Indore and Madurai.

इंदौर और मदुरै द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की औसत संख्या ज्ञात कीजिए।

- (A) 72
- (B) 84
- (C) 75
- (D) 69
- (E) 78

Q4: The total number of standard orders dispatched by Madurai is what percent less than the total number of standard orders dispatched by Indore?

मदुरै द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की कुल संख्या, इंदौर द्वारा भेजे गए स्टैंडर्ड ऑर्डरों की कुल संख्या से कितने प्रतिशत कम है?

- (A) 25% less
- (B) 26 2/3% less
- (C) 33 1/3% less
- (D) 20% less
- (E) 30% less

Q5: If the total orders dispatched by Pune hub are 25% more than the total orders dispatched by Madurai, and the standard orders dispatched by Pune are 25% more than the standard orders dispatched by Rajkot, then find the total number of express orders dispatched by Pune.

यदि पुणे केंद्र द्वारा भेजे गए कुल ऑर्डर, मदुरै द्वारा भेजे गए कुल ऑर्डरों से 25% अधिक हैं, और पुणे द्वारा भेजे गए स्टैंडर्ड ऑर्डर, राजकोट द्वारा भेजे गए स्टैंडर्ड ऑर्डरों से 25% अधिक हैं, तो पुणे द्वारा भेजे गए एक्सप्रेस ऑर्डरों की कुल संख्या ज्ञात कीजिए।

- (A) 84
- (B) 96
- (C) 90
- (D) 108

(E) 102

Solutions:**Basic calculation to find standard orders****On Indore hub**

Total orders dispatched = 138

Express orders dispatched = 48

Standard orders dispatched = Total orders - Express orders = $138 - 48 = 90$ **On Nashik hub**

Total orders dispatched = 114

Express orders dispatched = 42

Standard orders dispatched = Total orders - Express orders = $114 - 42 = 72$ **On Rajkot hub**

Total orders dispatched = 126

Express orders dispatched = 54

Standard orders dispatched = Total orders - Express orders = $126 - 54 = 72$ **On Madurai hub**

Total orders dispatched = 144

Express orders dispatched = 78

Standard orders dispatched = Total orders - Express orders = $144 - 78 = 66$ **On Guwahati hub**

Total orders dispatched = 144

Express orders dispatched = 36

Standard orders dispatched = Total orders - Express orders = $144 - 36 = 108$

Delivery Hub	Total Orders	Express Orders	Standard Orders
Indore	138	48	90
Nashik	114	42	72
Rajkot	126	54	72
Madurai	144	78	66
Guwahati	144	36	108

1. Option DStandard orders in Guwahati = $144 - 36 = 108$ Standard orders in Nashik = $114 - 42 = 72$ Difference = $108 - 72 = 36$ Required percentage = $36 \times 100 \div 72 = 50\%$ more

2. Option A

Standard orders in Indore = $138 - 48 = 90$

Standard orders in Rajkot = $126 - 54 = 72$

Required ratio = $90 : 72$

Required ratio = $5 : 4$

3. Option E

Standard orders in Indore = $138 - 48 = 90$

Standard orders in Madurai = $144 - 78 = 66$

Required average = $(90 + 66) \div 2 = 78$

4. Option B

Standard orders in Indore = 90

Standard orders in Madurai = 66

Difference = $90 - 66 = 24$

Required percentage = $24 \times 100 \div 90 = 26 \frac{2}{3}\%$ less

5. Option C

Total orders in Pune = $144 \times 125\% = 180$

Standard orders in Rajkot = $126 - 54 = 72$

Standard orders in Pune = $72 \times 125\% = 90$

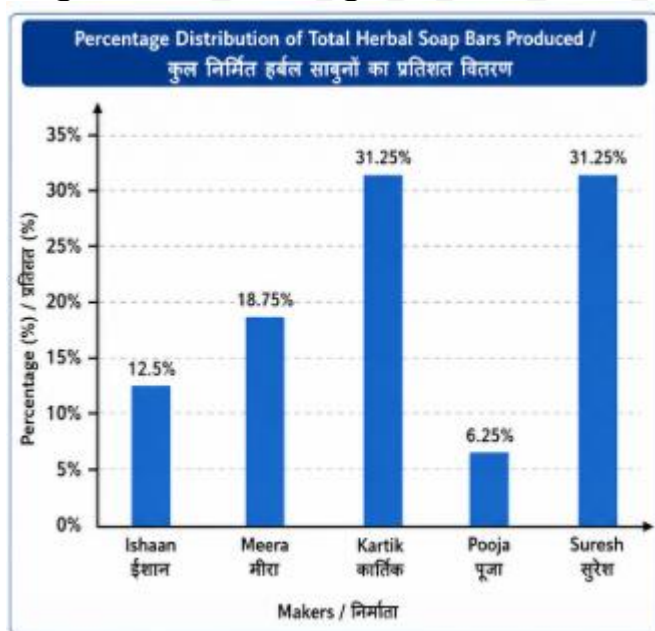
Express orders in Pune = $180 - 90 = 90$

Set 3: The bar graph shows the percentage distribution of total herbal soap bars produced by five different makers. The table shows the number of soap bars unsold by these makers. Total soap bars produced by all five makers together are 44,800.

बार ग्राफ पाँच अलग-अलग निर्माताओं द्वारा बनाए गए कुल हर्बल साबुनों का प्रतिशत वितरण दिखाता है। तालिका इन निर्माताओं के बिना बिके साबुनों की संख्या दिखाती है। सभी पाँच निर्माताओं द्वारा मिलाकर कुल 44,800 साबुन बनाए गए।

Note: Total soap bars produced by a particular maker = Soap bars sold + Soap bars unsold by that maker.

नोट: किसी विशेष निर्माता द्वारा बनाए गए कुल साबुन = उस निर्माता द्वारा बेचे गए साबुन + बिना बिके साबुन।



Unsold Soap Bars Table / बिना बिके साबुन तालिका	
Maker / निर्माता	Unsold Soap Bars / बिना बिके साबुन
Ishaan / ईशान	1540
Meera / मीरा	840
Kartik / कार्तिक	3500
Pooja / पूजा	700
Suresh / सुरेश	1260

Q1: Unsold soap bars of Ishaan and Meera together are how many more or less than the soap bars sold by Suresh?

ईशान और मीरा के मिलाकर बिना बिके साबुन, सुरेश द्वारा बेचे गए साबुनों से कितने अधिक या कम हैं?

- (A) 9590
- (B) 9870
- (C) 10360
- (D) 10710
- (E) 11200

Q2: If the manufacturing cost of one soap bar for Kartik is Rs. 24 and the selling price of one soap bar for Kartik is Rs. 30, then find the net profit or loss for Kartik. It is given that the unsold soap bars are thrown away.

यदि कार्तिक के लिए एक साबुन बनाने की लागत Rs. 24 है और एक साबुन का विक्रय मूल्य Rs. 30 है, तो कार्तिक का शुद्ध लाभ या हानि ज्ञात कीजिए। यह दिया गया है कि बिना बिके साबुन फेंक दिए जाते हैं।

- (A) 21000
- (B) 24500
- (C) 17500
- (D) 28000
- (E) 19600

Q3: What is the difference between the average number of soap bars sold by Ishaan, Meera, Kartik and Suresh, and the total unsold soap bars of all five makers?

ईशान, मीरा, कार्तिक और सुरेश द्वारा बेचे गए साबुनों की औसत संख्या और सभी पाँच निर्माताओं के कुल बिना बिके साबुनों के बीच का अंतर ज्ञात कीजिए।

- (A) 770
- (B) 945
- (C) 1015
- (D) 805
- (E) 875

Q4: What is the ratio of the total soap bars produced by Ishaan, Meera and Suresh together to the total soap bars sold by the same makers? If the ratio is $p:q$ in simplest form, find $p + q$.

ईशान, मीरा और सुरेश द्वारा मिलाकर बनाए गए कुल साबुनों का, उन्हीं निर्माताओं द्वारा मिलाकर बेचे गए कुल साबुनों से अनुपात ज्ञात कीजिए। यदि अनुपात सरलतम रूप में $p:q$ है, तो $p + q$ ज्ञात कीजिए।

- (A) 191
- (B) 187

- (C) 179
(D) 193
(E) 181

Q5: Use the serial numbers Ishaan = 1, Meera = 2, Kartik = 3, Pooja = 4 and Suresh = 5. The soap bars sold by which maker are equal to the total unsold soap bars of Meera and Suresh together? Find the serial number of that maker.

क्रम संख्या का उपयोग करें: ईशान = 1, मीरा = 2, कार्तिक = 3, पूजा = 4 और सुरेश = 5। किस निर्माता द्वारा बेचे गए साबुन, मीरा और सुरेश के मिलाकर कुल बिना बिके साबुनों के बराबर हैं? उस निर्माता की क्रम संख्या ज्ञात कीजिए।

- (A) 5
(B) 2
(C) 1
(D) 4
(E) 3

Solutions:

Maker / निर्माता	Produced / निर्मित	Unsold / बिना बिके	Sold / बेचे गए
Ishaan / ईशान	5600	1540	4060
Meera / मीरा	8400	840	7560
Kartik / कार्तिक	14000	3500	10500
Pooja / पूजा	2800	700	2100
Suresh / सुरेश	14000	1260	12740

1. Option C

Soap bars produced by Suresh = 14000

Unsold soap bars of Suresh = 1260

Soap bars sold by Suresh = $14000 - 1260 = 12740$

Unsold soap bars of Ishaan and Meera together = $1540 + 840 = 2380$

Difference = $12740 - 2380 = 10360$

So, Ishaan and Meera together are 10360 less than Suresh

2. Option A

Soap bars produced by Kartik = 14000

Unsold soap bars of Kartik = 3500

Soap bars sold by Kartik = $14000 - 3500 = 10500$

Total manufacturing cost = $14000 \times 24 = 336000$

Total selling amount = $10500 \times 30 = 315000$

Net loss = $336000 - 315000 = 21000$

3. Option E

Soap bars sold by Ishaan = $5600 - 1540 = 4060$

Soap bars sold by Meera = $8400 - 840 = 7560$

Soap bars sold by Kartik = $14000 - 3500 = 10500$

Soap bars sold by Suresh = $14000 - 1260 = 12740$

Average sold = $(4060 + 7560 + 10500 + 12740) \div 4 = 8715$

Total unsold soap bars = $1540 + 840 + 3500 + 700 + 1260 = 7840$

Difference = $8715 - 7840 = 875$

4. Option B

Total produced by Ishaan, Meera and Suresh = $5600 + 8400 + 14000 = 28000$

Total sold by Ishaan, Meera and Suresh = $4060 + 7560 + 12740 = 24360$

Required ratio = $28000:24360 = 100:87$

$p + q = 100 + 87 = 187$

5. Option D

Unsold soap bars of Meera and Suresh together = $840 + 1260 = 2100$

Soap bars sold by Pooja = $2800 - 700 = 2100$

Required maker = Pooja

Serial number of Pooja = 4

Set 4: A survey was conducted among 1312 members of a community learning centre to study their enrollment in three weekend workshops: Coding, Graphic Design and Financial Planning. The ratio of members enrolled only in Coding, only in Graphic Design and only in Financial Planning is 5 : 6 : 4, respectively. The number of members enrolled in only Coding and Financial Planning is 75% of the number of members enrolled in only Coding and Graphic Design. The number of members enrolled in all three workshops is 50% of the number of members enrolled only in Graphic Design. The number of members enrolled in only Graphic Design and Financial Planning is 32 more than the number of members enrolled only in Financial Planning. The ratio of members enrolled only in Coding to those enrolled in only Coding and Financial Planning is 5 : 2.

एक सामुदायिक शिक्षण केंद्र के 1312 सदस्यों पर एक सर्वे किया गया, ताकि तीन सप्ताहांत वर्कशॉप: कोडिंग, ग्राफिक डिजाइन और वित्तीय योजना में उनके नामांकन का अध्ययन किया जा सके। केवल कोडिंग, केवल ग्राफिक डिजाइन और केवल वित्तीय योजना में नामांकित सदस्यों का अनुपात क्रमशः 5 : 6 : 4 है। केवल कोडिंग और वित्तीय योजना में नामांकित सदस्यों की संख्या, केवल कोडिंग और ग्राफिक डिजाइन में नामांकित सदस्यों की संख्या का 75% है। तीनों वर्कशॉप में नामांकित सदस्यों की संख्या, केवल ग्राफिक डिजाइन में नामांकित सदस्यों की संख्या का 50% है। केवल ग्राफिक डिजाइन और वित्तीय योजना में नामांकित सदस्यों की संख्या, केवल वित्तीय योजना में नामांकित सदस्यों की संख्या से 32 अधिक है। केवल कोडिंग में नामांकित सदस्यों और केवल कोडिंग तथा वित्तीय योजना में नामांकित सदस्यों का अनुपात 5 : 2 है।

Q1: Find the number of members who are enrolled in Graphic Design.
ग्राफिक डिजाइन में नामांकित सदस्यों की संख्या ज्ञात कीजिए।

- (A) 744
- (B) 768
- (C) 816
- (D) 784
- (E) 804

Q2: The number of members enrolled only in Financial Planning is $p\%$ of the number of members enrolled only in Coding and Graphic Design. Find the value of p .

केवल वित्तीय योजना में नामांकित सदस्यों की संख्या, केवल कोडिंग और ग्राफिक डिजाइन में नामांकित सदस्यों की संख्या का $p\%$ है। p का मान ज्ञात कीजिए।

- (A) 150

- (B) 125
- (C) 175
- (D) 140
- (E) 160

Q3: Find the difference between the number of members enrolled in at most one workshop and the number of members enrolled in all three workshops.

अधिकतम एक वर्कशॉप में नामांकित सदस्यों की संख्या और तीनों वर्कशॉप में नामांकित सदस्यों की संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 528
- (B) 552
- (C) 600
- (D) 624
- (E) 576

Q4: If the simplified ratio of members enrolled in exactly one workshop to members enrolled in exactly two workshops is $a : b$, find the value of $a + b$.

यदि ठीक एक वर्कशॉप में नामांकित सदस्यों और ठीक दो वर्कशॉप में नामांकित सदस्यों का सरलतम अनुपात $a : b$ है, तो $a + b$ का मान ज्ञात कीजिए।

- (A) 71
- (B) 73
- (C) 75
- (D) 77
- (E) 79

Q5: Find the number of members who are enrolled in at least two workshops.

कम से कम दो वर्कशॉप में नामांकित सदस्यों की संख्या ज्ञात कीजिए।

- (A) 568
- (B) 584
- (C) 592
- (D) 608
- (E) 616

Solutions:

Region क्षेत्र	Number of Members सदस्यों की संख्या
Only Coding केवल कोडिंग	240
Only Graphic Design केवल ग्राफिक डिजाइन	288
Only Financial Planning केवल वित्तीय योजना	192
Only Coding and Graphic Design केवल कोडिंग और ग्राफिक डिजाइन	128
Only Coding and Financial Planning केवल कोडिंग और वित्तीय योजना	96
Only Graphic Design and Financial Planning केवल ग्राफिक डिजाइन और वित्तीय योजना	224
All three workshops तीनों वर्कशॉप	144
Total कुल	1312

1. Option D

Members enrolled in Graphic Design = Only Graphic Design + Only Coding and Graphic Design + Only Graphic Design and Financial Planning + All three
 $= 288 + 128 + 224 + 144$
 $= 784$

2. Option A

Only Financial Planning = 192
 Only Coding and Graphic Design = 128
 $p = 192 \div 128 \times 100$
 $= 150$

3. Option E

Members enrolled in at most one workshop = $240 + 288 + 192$
 $= 720$
 Members enrolled in all three workshops = 144
 Required difference = $720 - 144$
 $= 576$

4. Option B

Members enrolled in exactly one workshop = $240 + 288 + 192$
 $= 720$

Members enrolled in exactly two workshops = $128 + 96 + 224$

= 448

Required ratio = $720 : 448$

= $45 : 28$

$a + b = 45 + 28$

= 73

5. Option C

Members enrolled in exactly two workshops = $128 + 96 + 224$

= 448

Members enrolled in all three workshops = 144

Members enrolled in at least two workshops = $448 + 144$

= 592